

EEE4114F Class Test 2

May 2023

Duration: 1 hours

Total: 40 marks

Empl ID: _____

Instructions:

- This is a closed-book test but you may use a calculator.
- There are two parts to this test.
- **Part A** has *two* questions totalling 20 marks. You must answer all of them.
- **Part B** has *two* questions totalling 20 marks. You must answer all of them.
- Part A and part B each count 50% of the final mark.
- You must use a pen to complete the exam — pencil will only be marked for drawings. Values indicated on drawings must be written in pen. Please show all of your working clearly — method and approach matter.
- A table of standard Fourier transform and z-transform pairs appears at the end of this paper.
- You have 1 hour.

Part A:	Question:	1	2	Total
	Points:	10	10	20
	Score:			

Part B:	Question:	3	4	Total
	Points:	10	10	20
	Score:			

Part A: Digital Signal Processing

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Question 1 [10 marks]

The impulse response of a discrete-time LTI system is

$$h[n] = -\delta[n] + 2\left(\frac{1}{2}\right)^n u[n].$$

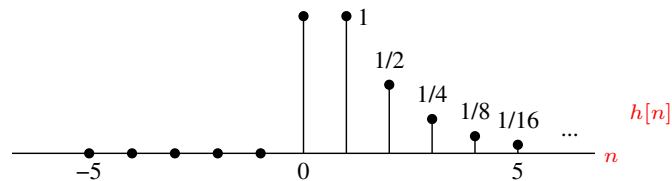
- Plot $h[n]$ over the range $n = -5$ to $n = 5$.
- Is the system causal? Why?
- Draw a pole-zero plot of the system function $H(z)$.
- Assuming that

$$H(z) = \frac{1 + 1/2z^{-1}}{1 - 1/2z^{-1}} \quad \text{for } |z| > 1/2,$$

or otherwise, find the impulse response $g[n]$ of the stable inverse of the system.

Solution:

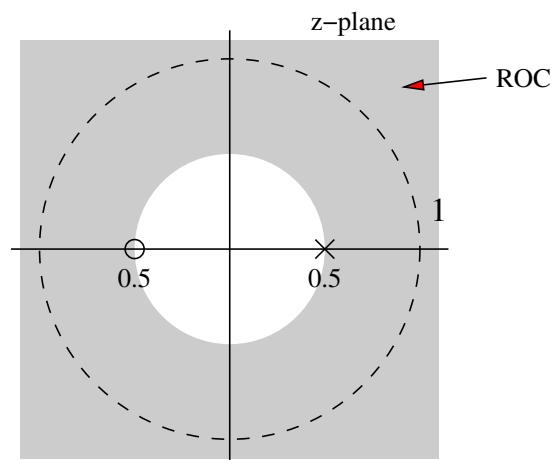
- Plot as follows:



- The system is causal because the impulse response is right sided.
- The system function is

$$H(z) = -1 + 2 \frac{1}{1 - 1/2z^{-1}} = \frac{-(1 - 1/2z^{-1}) + 2}{1 - 1/2z^{-1}} = \frac{1 + 1/2z^{-1}}{1 - 1/2z^{-1}} = \frac{z + 1/2}{z - 1/2}$$

with ROC $|z| > 1/2$, so the pole-zero plot is as follows:

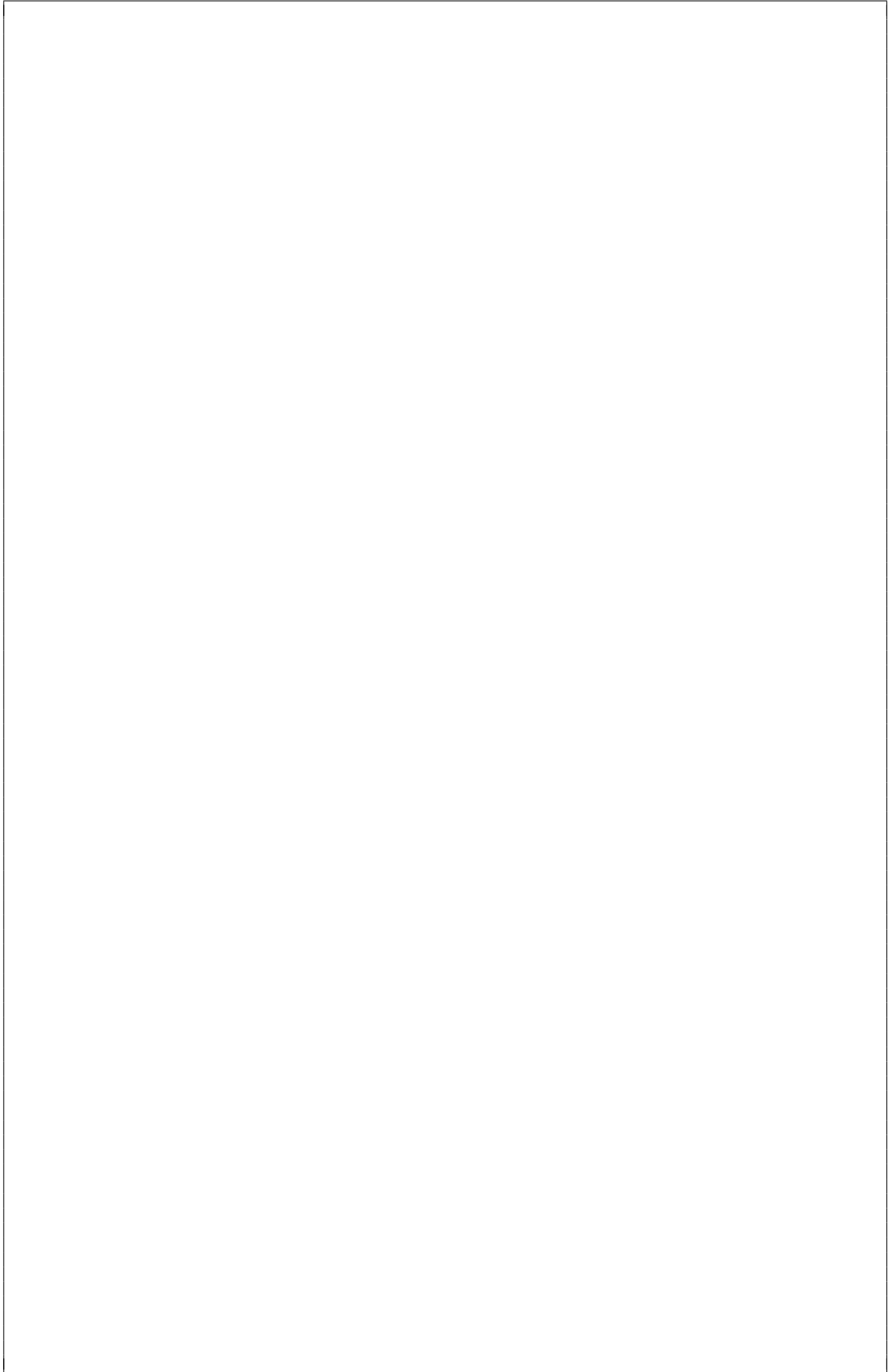


d. The stable (and causal) inverse system has z-transform

$$G(z) = \frac{1 - 1/2z^{-1}}{1 + 1/2z^{-1}} \quad \text{for } |z| > 1/2,$$

which has impulse response

$$g[n] = (-1/2)^n u[n] - 1/2(-1/2)^{n-1} u[n-1].$$



Question 2 [10 marks]

- a. The continuous-time signal $x_c(t) = \cos(40\pi t)$ is sampled with a sampling period T to obtain the discrete-time signal $x[n] = \cos(\pi n/15)$. Determine a choice for T consistent with this information.
- b. Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.
- c. The continuous-time signal

$$x_c(t) = \frac{\cos(40\pi t)}{160\pi t}$$

is sampled with a sampling period T to obtain the discrete-time signal

$$x[n] = \frac{\cos(\pi n/15)}{4\pi n/15}.$$

Determine a choice for T consistent with this information.

- d. Is your choice for T above unique? If so, explain why. If not, specify another choice of T consistent with the information given.

Solution:

- a. In general we sample according to $t = nT$, so the sampled signal is $x_c(nT) = \cos(40\pi nT)$. The signals are equal whenever $\cos(40\pi nT) = \cos(\pi/15n)$, or when

$$40\pi nT = \pi/15 + 2\pi k \quad \text{or} \quad T = 1/600 + 1/20k$$

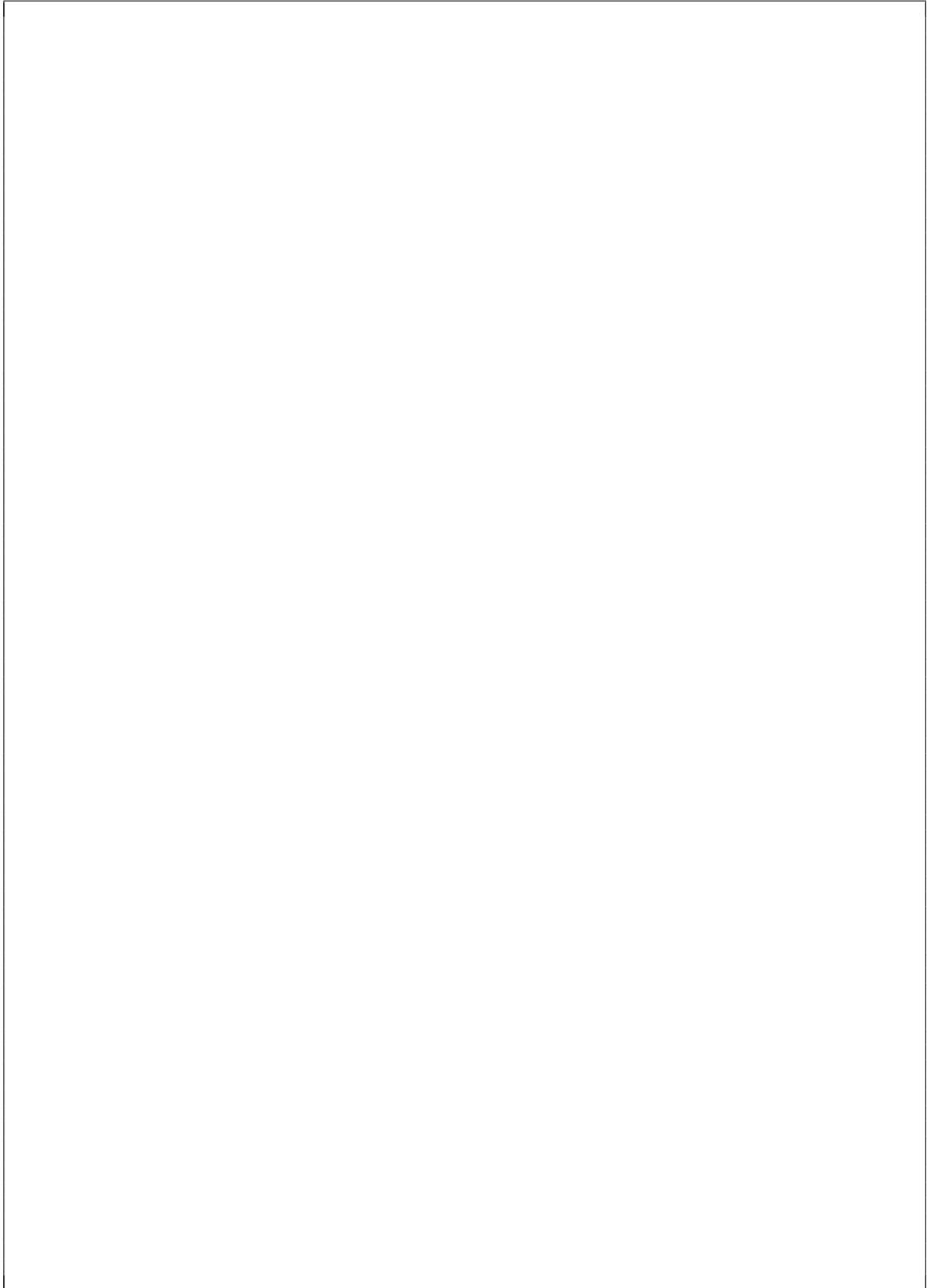
for any integer k . The case of $k = 0$ yields one solution $T = 1/600$ seconds.

- b. Any other value of k yields a different solution, so there are clearly infinitely many.
- c. By inspection, with $t = nT$ and $T = 1/600$ we have

$$x_c(nT) = 1/4 \frac{\cos(40\pi nT)}{40\pi nT} = 1/4 \frac{\cos(\pi/15n)}{\pi/15n} = x[n].$$

Thus $T = 1/600$ seconds provides a consistent solution.

- d. Since $x_c(t)$ is not periodic the solution in the previous instance is unique.



END OF PART A

Part B: Machine Learning

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Question 3 [10 marks]

Suppose you have an unknown sensor that is used to measure the distance to an object. The sensor produces voltages as outputs that you can measure. However, you do not know how to convert the voltages to distances. Figure 1. shows all of the collected training data and the results (A, B, C) of training a linear regression model three times using different training data randomly sampled from the full dataset.

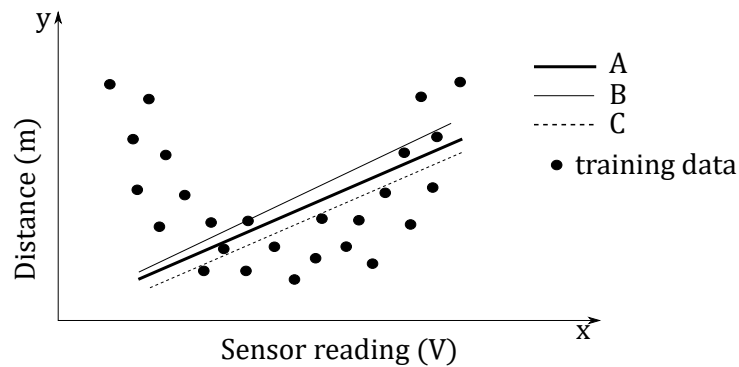


Figure 1: The output of a linear regression model trained three times with different samples of the training data

- a. Would you consider this model to have a high or low bias? Explain your answer. (2)

Solution: High bias, as the model is unable to learn to predict the training data with a reasonable amount of accuracy. It appears to have an inappropriate form for the task.

- b. Would you consider this model to have a high or low variance? Explain your answer. (2)

Solution: Low variance, as the model appears to learn similar parameters across the multiple optimizations on different samples of the data.

At this point, you decide to train a polynomial regression model with the model's output shown in Figure 2 when trained on three different splits of the training data.

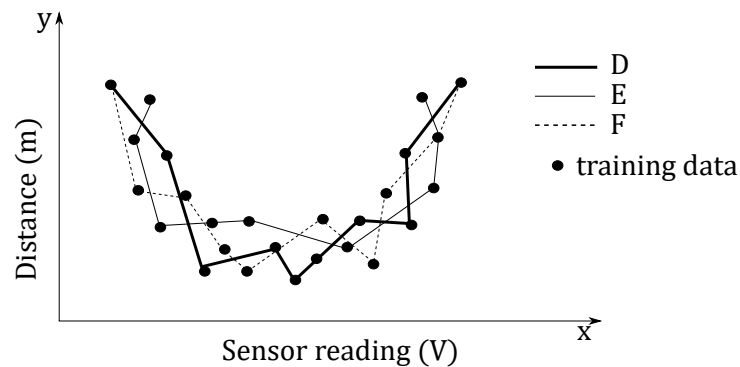


Figure 2: The output of a polynomial regression model trained three times with different samples of the training data

- c. Would you consider this model to have a high or low bias? Explain your answer. (2)

Solution: Low bias, as the model has the capacity to learn to predict the training data sufficiently

- d. Would you consider this model to have a high or low variance? Explain your answer. (2)

Solution: High variance, as across the multiple optimizations on different samples of the data, the model appears to learn very different parameters that produce different outputs depending on the training data that was used.

- e. Which of the models (linear or polynomial) would benefit the most from regularization? Explain your answer. (2)

Solution: The polynomial regression model would benefit the most as it appears to have a large number of parameters based on its flexibility. Regularization tries to limit the effects of model parameters and often results in parameters being reduced to zero, or close to it. This effectively simplifies the behaviour of the model, and in the case of model 2, could alleviate the clear overfitting problems that are shown.

Question 4 [10 marks]

Suppose you collect a data set for the previous distance measuring sensor, $\mathcal{D}$, consisting of input-output pairs (x_i, y_i) where x_i is the measurement from the sensor in, V, and y_i is the target variable, which in this case is the distance from the sensor to a wall in meters.

$$\mathcal{D} = \{(0.2, 3), (0.5, 2), (1.3, 1.3), (2, 4)\}$$

You then decide to fit a regression model of the form $h_\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$ to it using gradient descent optimization methods. The model's parameters are randomly initialised such that

$$\theta_0 = 1, \theta_1 = 2, \theta_2 = 6$$

The objective function, J , you choose to optimize is given by the mean square error of the form:

$$J = \frac{1}{2N} \sum_{i=1}^N (h_\theta(x_i) - y_i)^2$$

Where N is the number of examples in a single training iteration.

Parameters are updated after **each** iteration, and a single iteration uses one example from the data set. The update rule is given by:

$$\theta_i := \theta_i - \eta \frac{\partial J}{\partial \theta_i}$$

Where $\eta = 0.1$. Answer the following questions

- a. Determine the expressions for the gradients of the objective function with respect to the model's parameters: $\frac{\partial J}{\partial \theta_0}$, $\frac{\partial J}{\partial \theta_1}$, and $\frac{\partial J}{\partial \theta_2}$. (3)

Solution:

$$\begin{aligned} \frac{\partial J}{\partial \theta_0} &= \frac{2}{2} \left(\sum_{i=1}^1 ((\theta_0 + \theta_1 x + \theta_2 x^2) - y_i) \right) = (\theta_0 + \theta_1 x + \theta_2 x^2) - y_i \\ \frac{\partial J}{\partial \theta_1} &= \frac{2}{2} \left(\sum_{i=1}^1 ((\theta_0 + \theta_1 x + \theta_2 x^2) - y_i) x_i \right) = ((\theta_0 + \theta_1 x + \theta_2 x^2) - y_i) x_i \\ \frac{\partial J}{\partial \theta_2} &= \frac{2}{2} \left(\sum_{i=1}^1 ((\theta_0 + \theta_1 x + \theta_2 x^2) - y_i) x_i^2 \right) = ((\theta_0 + \theta_1 x + \theta_2 x^2) - y_i) x_i^2 \end{aligned}$$

- b. Calculate the gradients of the parameters after one training iteration using the first training pair in the data set $(x_1, y_1) = (0.2, 3)$. Round your answers to two decimal places. (3)

Solution:

$$h_{\theta}(0.2) = 1 + 2(0.2) + 6(0.2^2) = 1.64$$

$$\frac{\partial J}{\partial \theta_0} = (1.64 - 3) = -1.36$$

$$\frac{\partial J}{\partial \theta_1} = (1.28 - 3)(0.2) = -0.272 = -0.27$$

$$\frac{\partial J}{\partial \theta_2} = (1.28 - 3)(0.2)^2 = -0.0544 = -0.05$$

- c. Calculate the values for θ_0 and θ_2 after the second training iteration where $(x_2, y_2) = (0.5, 2)$. You can assume the model output is given by $h_{\theta}(0.5) = 3.7$. (4)

Solution: First we need to know the parameters after the first update.

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 1 - 0.1(-1.36) \\ &:= 1.14\end{aligned}$$

$$\begin{aligned}\theta_2 &:= \theta_2 - \eta \frac{\partial J}{\partial \theta_2} \\ &:= 6 - 0.1(-0.05) \\ &:= 6.01\end{aligned}$$

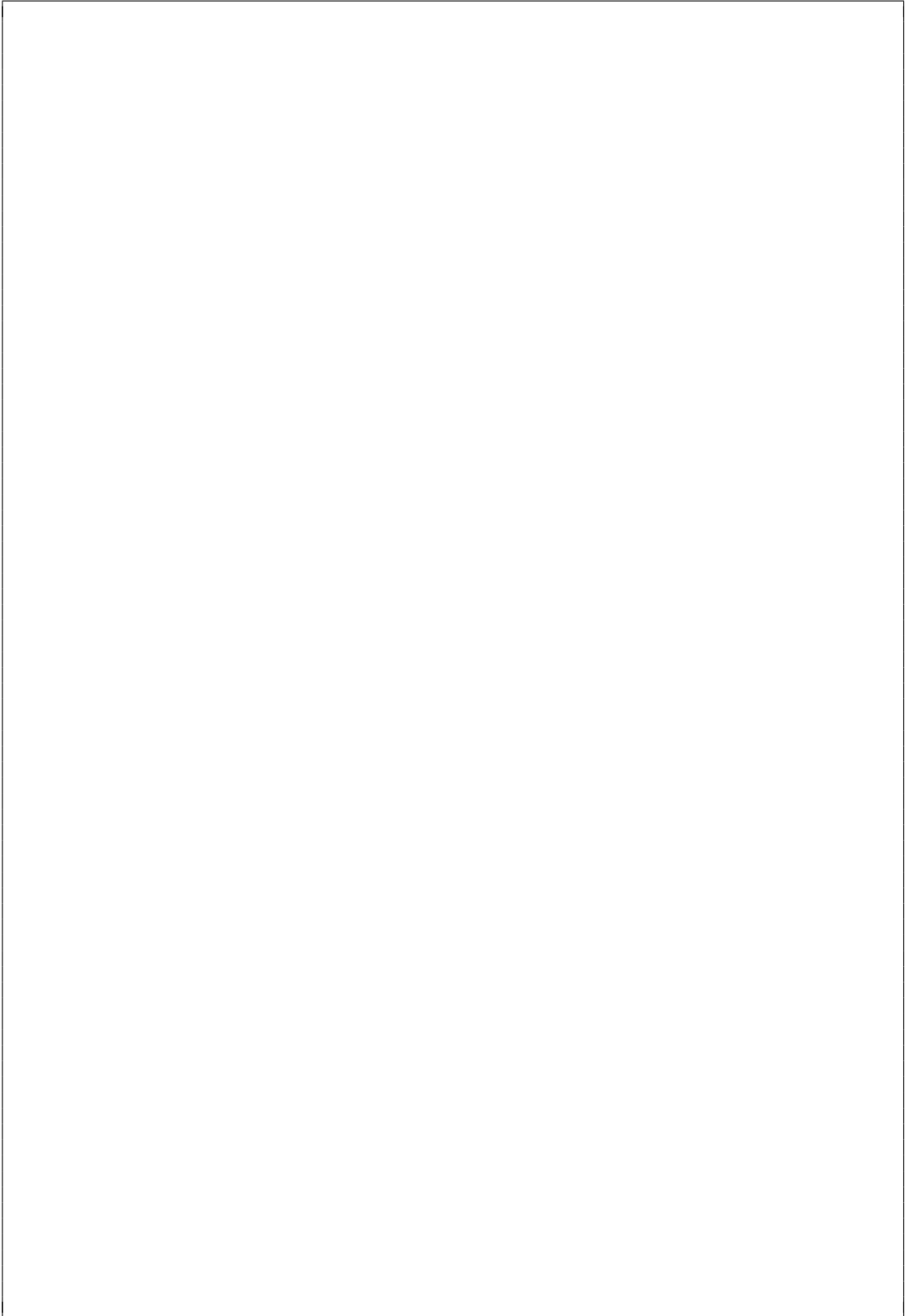
Then repeat the calculation of gradients, but using the new parameters and new data $h_{\theta}(x_i) = 1.14 + 2.03(x_i) + 6.01(x_i^2)$. Where the next example is $(0.5, 2)$. You are given $h_{\theta}(0.5) = 3.7$, so it is unnecessary to find θ_1 .

$$\begin{aligned}\frac{\partial J}{\partial \theta_0} &= (3.7 - 2) = 1.7 \\ \frac{\partial J}{\partial \theta_1} &= (3.7 - 2)(0.5) = 0.425\end{aligned}$$

Then the new parameters are

$$\begin{aligned}\theta_0 &:= \theta_0 - \eta \frac{\partial J}{\partial \theta_0} \\ &:= 1.14 - 0.1(1.7) \\ &:= 0.97\end{aligned}$$

$$\begin{aligned}\theta_2 &:= \theta_2 - \eta \frac{\partial J}{\partial \theta_2} \\ &:= 6.01 - 0.1(0.43) \\ &:= 5.97\end{aligned}$$



END OF PART B

DSP information sheet

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t} dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t} d\Omega \quad (\text{CTFT})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad (\text{DTFT})$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j} \oint_C X(z)z^{n-1} dz \quad (\text{ZT})$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with } W_N = e^{-j\frac{2\pi}{N}} \quad (\text{DFT})$$

Discrete-time Fourier transform properties

Sequences $x[n], y[n]$	Transforms $X(e^{j\omega}), Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d} X(e^{j\omega})$	Time shift
$e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
$u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \omega < \omega_c \\ 0 & \omega_c < \omega \leq \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \leq n \leq M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n], y[n]$	Transforms $X(z), Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC = R_x	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC = $ z_0 R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC = $\frac{1}{R_x}$	Time reversal
$nx[n]$	$-z \frac{dX(z)}{dz}$	ROC = R_x	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC = R_x	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$ z < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
$\begin{cases} a^n & 0 \leq n \leq N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$ z > 0$
$\cos(\omega_0 n) u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$ z > 1$
$r^n \cos(\omega_0 n) u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$ z > r $

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1-a^N}{1-a} \quad \text{for } a \neq 1.$$